Mixed Angle Relationships and Reasoning

Year 6 · Mathematics · Measurement | C5 · Learn + Notice

Name: _____ Date: _____

LEARNING INTENTION

I can choose a rule, calculate x and check the result.

KEY IDEA

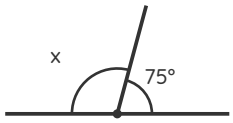
Identify the relationship first: right angle 90°, straight angle 180°, full turn 360°, or equal vertically opposite angles.

VOCABULARY

relationship · equation · reasoning · check

LEARN IT: A Worked Model

For a known 75° angle and its adjacent angle on a straight line, use 180°, not 360°. Write x = 180° - 75° = 105°.



Use the marked opening. Diagrams for calculation tasks are not drawn to scale.

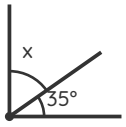
SUCCESS CRITERIA

- ☐ I can choose a rule, calculate x and check the result.
- ☐ I can explain the benchmark, tool or angle rule I used.
- ☐ I can check my answer and correct a mistake.

NOTICE IT: Guided Practice

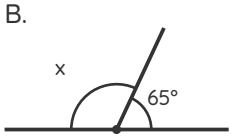
Find x. Write the rule or total you used.

A.



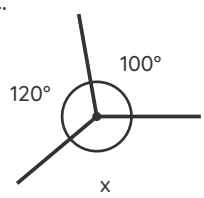
Answer: _____

B.



Answer: _____

C.



Answer: _____

© 2026 Aussie Primary Academy. All Rights Reserved.

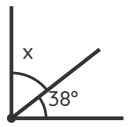
Mixed Angle Relationships and Reasoning

Year 6 · Mathematics · Measurement | C5 · Practise + Apply + Reason

PRACTISE IT: Independent Questions

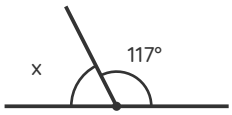
Find x. Write the rule or total you used.

A.



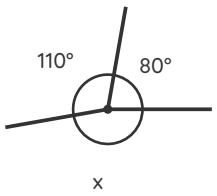
Answer: _____

B.



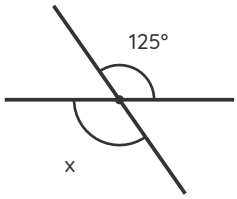
Answer: _____

C.



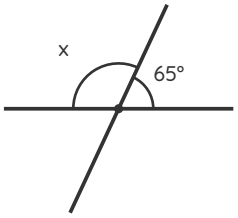
Answer: _____

D.



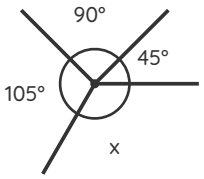
Answer: _____

E.



Answer: _____

F.



Answer: _____

APPLY IT

Two straight lines cross. One angle is 42°. Find all four angles and check that their total is 360°.

EXPLAIN HOW YOU KNOW: Misconception Check

A student writes x = 360° - 117° for practice B. Explain the mistake and correct the equation.

CHALLENGE: Reasoning Task

Four angles around a point are x, x, 80° and 100°. Find x. Show your equation.

MY SELF-REVIEW

Circle one: I understand it / I need more practice / I need help

© 2026 Aussie Primary Academy. All Rights Reserved.

Mixed Angle Relationships and Reasoning

Year 6 · Mathematics · Measurement | C5 · Teacher Copy

ANSWER KEY: Guided + Independent Practice

Notice: A 55°; B 115°; C 140°.

Practice: A 52° (right); B 63° (straight); C 170° (point); D 125° (opposite); E 115° (adjacent straight); F 120° (point).

Apply: 42°, 138°, 42°, 138°; total 360°.

Reason: B fills a straight angle, so x = 180° - 117° = 63°.

Challenge: 2x + 80° + 100° = 360°; x = 90°.

COMMON MISCONCEPTION

Do not decide the rule from diagram appearance alone. Identify the outer arms and marked opening.

TEACHER PROMPT

Ask: "Which opening are you using? What benchmark or angle relationship applies? How can you check your answer?"

NEXT-STEP GUIDANCE

Consolidate any rule that was not explained confidently before extending to Year 7.

PRINTING AND ASSESSMENT

Print A4. Diagrams are not to scale for calculation purposes. Use stated values and angle relationships; check the explanation as well as the result.

© 2026 Aussie Primary Academy. All Rights Reserved.